

Q1 - LINUX COMMANDS BASICS

1. Create and navigate to project directory

```
Harshit@Harshit MINGW64 ~  
$ mkdir Harshit-GLA-CourseRegistration  
  
Harshit@Harshit MINGW64 ~  
$ cd Harshit-GLA-CourseRegistration  
  
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ pwd  
/c/Users/Harshit/Harshit-GLA-CourseRegistration
```

2. Create project structure

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ mkdir -p src docs backup  
  
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ touch src/course_search.txt course_info.txt docs/README.md backup/backup_info.txt
```

3. List all files and directories

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ ls -R  
.:  
backup/   course_info.txt  docs/   src/  
  
./backup:  
backup_info.txt  
  
./docs:  
README.md  
./src:  
course_search.txt  
  
3 directories, 4 files
```

4. Check current location and contents (detailed)

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ ls -la  
total 5  
drwxr-xr-x 1 Harshit 197609  0 Sep 02 10:15 .  
drwxr-xr-x 1 Harshit 197609  0 Sep 02 10:10 ..  
drwxr-xr-x 1 Harshit 197609  0 Sep 02 10:15 backup  
-rw-r--r-- 1 Harshit 197609  0 Sep 02 10:15 course_info.txt  
drwxr-xr-x 1 Harshit 197609  0 Sep 02 10:15 docs  
drwxr-xr-x 1 Harshit 197609  0 Sep 02 10:15 src  
total 5  
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration  
$ █
```

Q4 - MERGING & CONFLICT RESOLUTION

PART A - Merge course-search

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git merge course-search

Updating cdfdaeb..c378831

Fast-forward

src/course_search.txt | 4 +++
1 file changed, 4 insertions(+)
create mode 100644 src/course_search.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ ls src

course_search.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git log --oneline --all --graph

* c378831 (HEAD -> master, course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup

----- PART B - Generate conflict and resolve -----

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git branch course-update

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git switch course-update

Switched to branch 'course-update'

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (course-update)

\$ nano course_info.txt # (edited the line: Project Name: Course Registration System)

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (course-update)

\$ git add course_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (course-update)

\$ git commit -m "Update project name in course-update"

[course-update f5b3a72] Update project name in course-update
1 file changed, 1 insertion(+), 1 deletion(-)

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (course-update)

\$ git switch master

Switched to branch 'master'

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ nano course_info.txt # (edited the line differently: Project Name: Online Course Registration System)

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git add course_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git commit -m "Update project name in master"

[master 3e8d1b9] Update project name in master
1 file changed, 1 insertion(+), 1 deletion(-)

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git merge course-update

Auto-merging course_info.txt

CONFLICT (content): Merge conflict in course_info.txt

Automatic merge failed; fix conflicts and then commit the result.

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master|MERGING)

\$ git status

On branch master

You have unmerged paths.

(fix conflicts and run "git commit")

Unmerged paths:

(use "git add <file>..." to mark resolution)

both modified: course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master|MERGING)

\$ nano course_info.txt # (manually resolved the conflict)

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master|MERGING)

\$ cat course_info.txt

Project Name: Online Course Registration System

Course Duration: 4 Years

Created By: Harshit Arora

Last Updated: 02-09-2026

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master|MERGING)

\$ git add course_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master|MERGING)

\$ git commit -m "Resolve merge conflict in course information"

[master a7d9c41] Resolve merge conflict in course information

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git status

On branch master

nothing to commit, working tree clean

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

\$ git log --oneline --all --graph

* a7d9c41 (HEAD -> master) Resolve merge conflict in course information

| \

| * f5b3a72 (course-update) Update project name in course-update

* | 3e8d1b9 Update project name in master

| /

* c378831 (course-search) Add course search module

* cdfdaeb Update course information

* b45e4c5 Initial project setup

```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
##### Q5 - FINAL DEMONSTRATION #####
1. Complete Linux project structure
$ tree
.
|-- backup
|-- docs
|-- src
|   |-- course_search.txt
|-- course_info.txt

3 directories, 2 files
2. Git repository status
$ git status
On branch master
nothing to commit, working tree clean

3. Git history
$ git log --oneline --all --graph
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
| \
| * f5b3a72 (course-update) Update project name in course-update
| * | 3e8d1b9 Update project name in master
| /
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup

4. Feature branches
$ git branch
  course-search
  course-update
* master

5. Successful feature merge (Final status)
$ git status
On branch master
nothing to commit, working tree clean

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ |

```

Q2 - GIT REPOSITORY INITIALIZATION

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git init
Initialized empty Git repository in C:/Users/Harshit/Harshit-GLA-CourseRegistration/.git/

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git config user.name "Harshit Arora"

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git config user.email "harshit.arora@example.com"

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git add .

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git commit -m "Initial project setup"
[master b45e4c5] Initial project setup
 2 files changed, 11 insertions(+)
 create mode 100644 course_info.txt
 create mode 100644 docs/README.md

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git status
On branch master
nothing to commit, working tree clean

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ git log --oneline
b45e4c5 (HEAD -> master) Initial project setup

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)
$ █
```


Q3 - GIT BRANCHING AND FILE UPDATES

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ git branch
  course-search
* master
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ ls
backup/  course_info.txt  docs/  src/
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ nano course_info.txt
# (updated the Project Name line)
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ git add course_info.txt
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ git commit -m "Update course information"
[master cdfdaeb] Update course information
1 file changed, 1 insertion(+), 1 deletion(-)
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ git log --oneline --all --graph
* cdfdaeb (HEAD -> master) Update course information
* b45e4c5 Initial project setup
```

Harshit@Harshit MINGW64 ~/Harshit-GLA-CourseRegistration (master)

```
$ cat course_info.txt
Project Name: Task Scheduler with Priority Scheduling
Course Duration: 4 Years
Created By: Harshit Arora
Last Updated: 02-09-2026
```